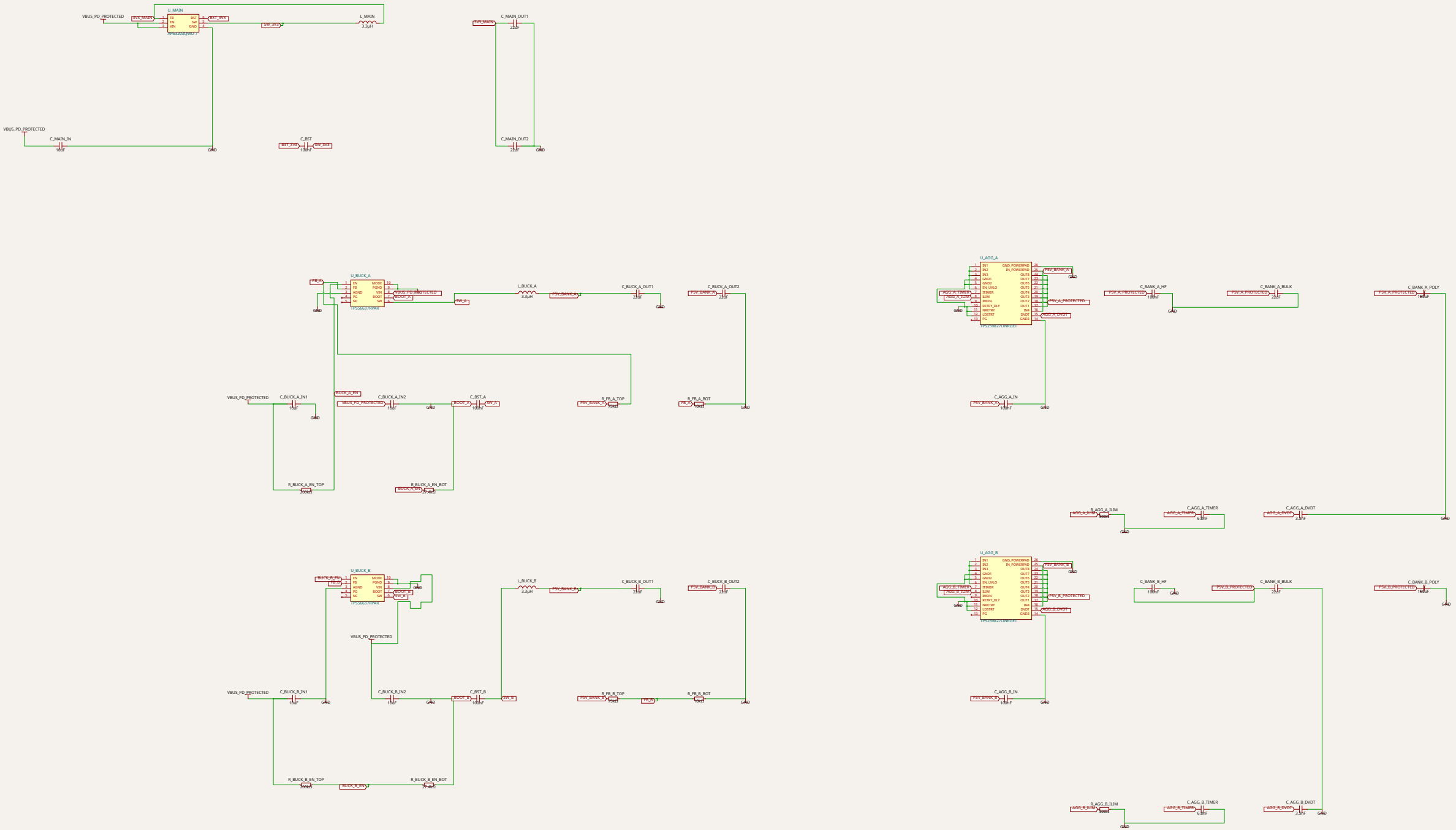


USB_CONTROLLED_DEBUG_HUB_2A

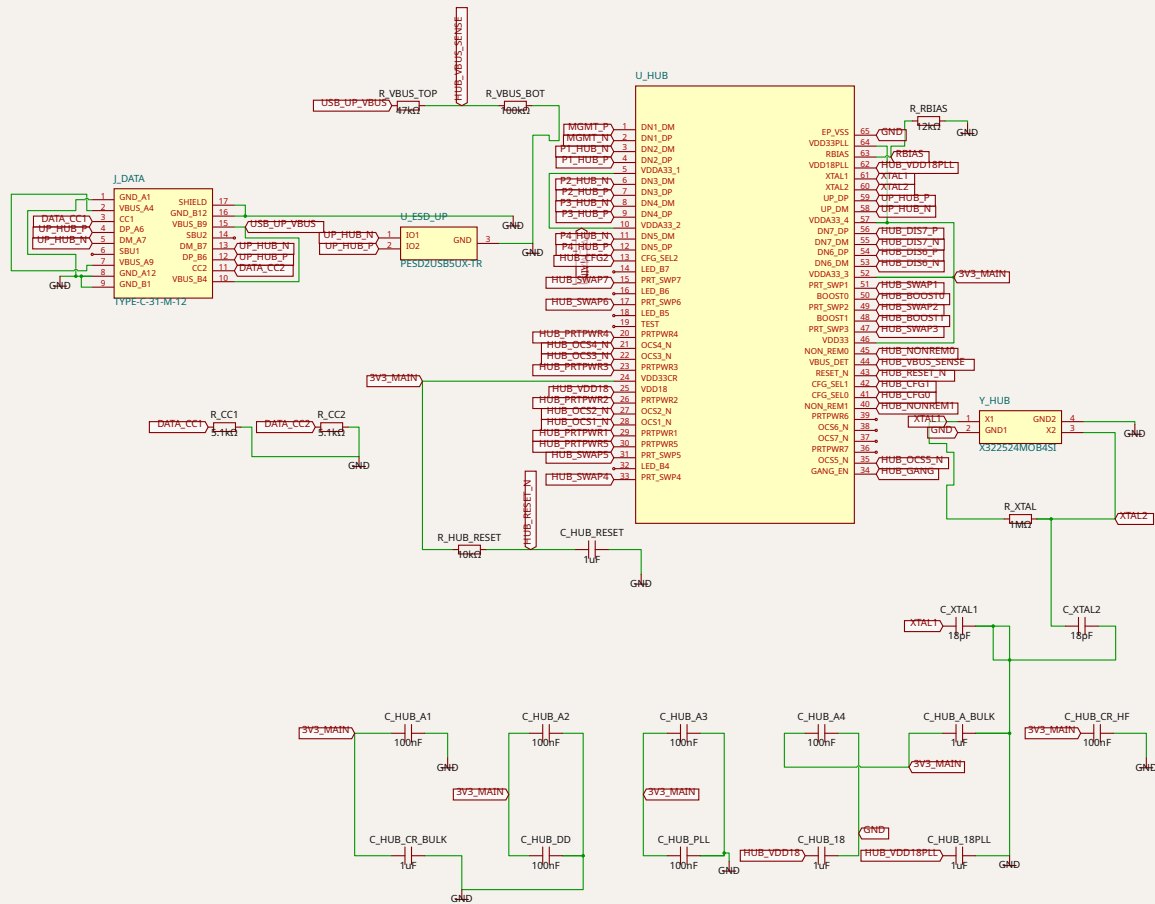
POWER DISTRIBUTION — dual 5 V / 6 A banks, bank eFuses and 3.3 V buck

Page 2 of 10 • exact circuit.json SHA-256 91fa12ab65d3eac8... • 44 components • landscape page fit



USB_CONTROLLED_DEBUG_HUB_2A

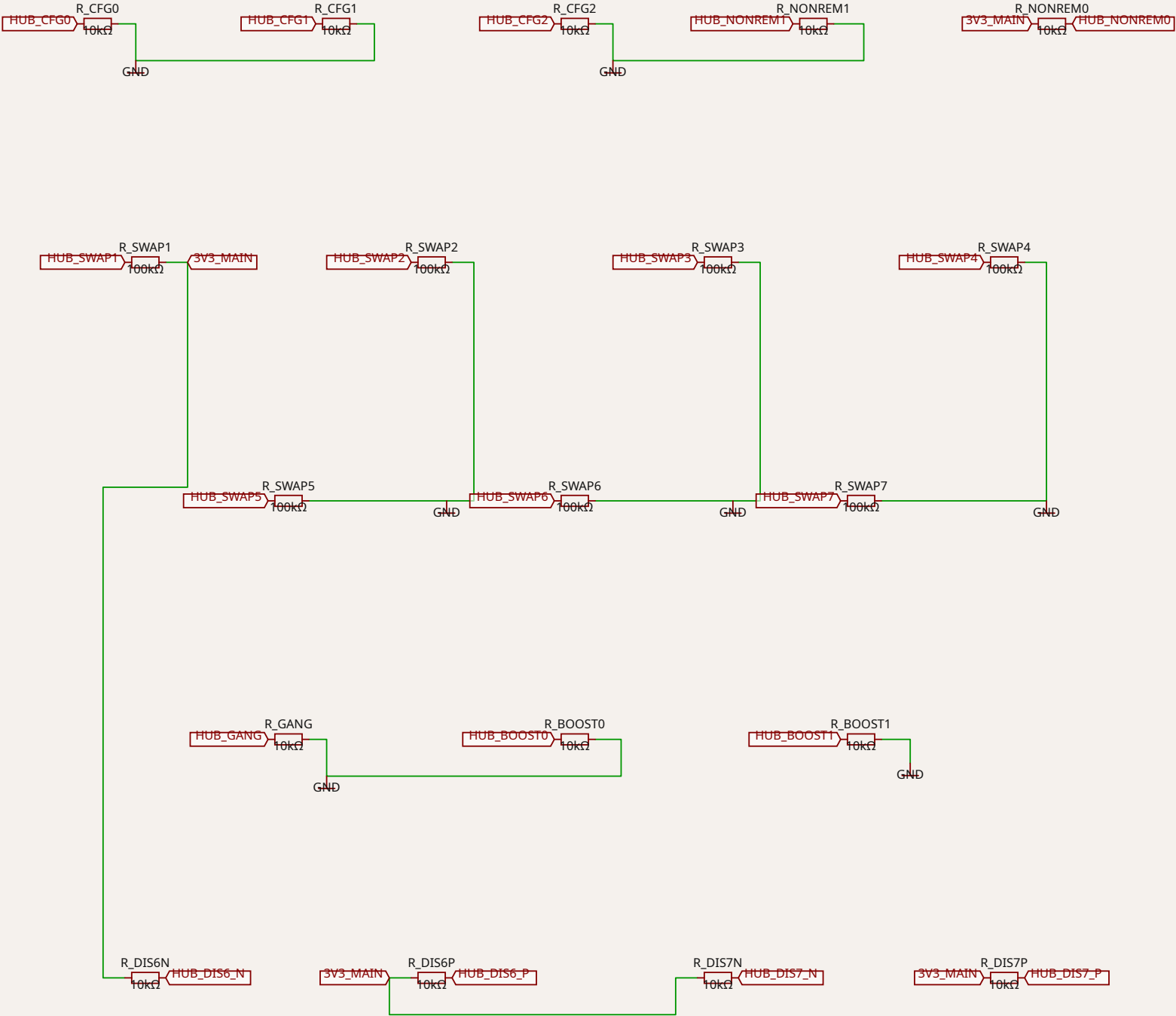
USB HUB — USB-C data-only upstream, ESD, USB2517I, straps, clock and bypass



INTENTIONAL NC: U_HUB LED/TEST and disabled-port 6/7 PWR/OCS pins.

USB_CONTROLLED_DEBUG_HUB_2A

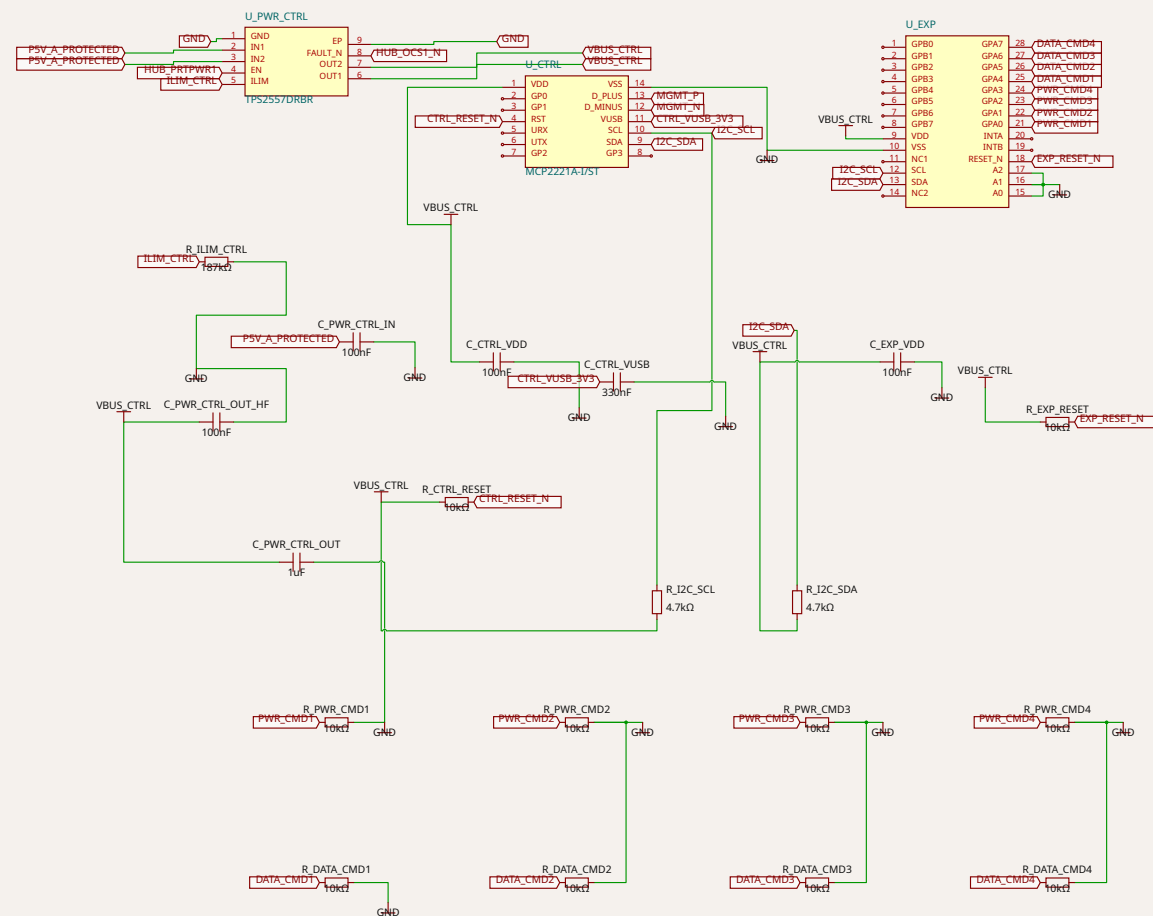
USB HUB CONFIGURATION — P1 swapped, P2-5 normal, P6-7 disabled



USB_CONTROLLED_DEBUG_HUB_2A

MANAGEMENT — factory MCP2221A HID/I2C and MCP23017 command bank

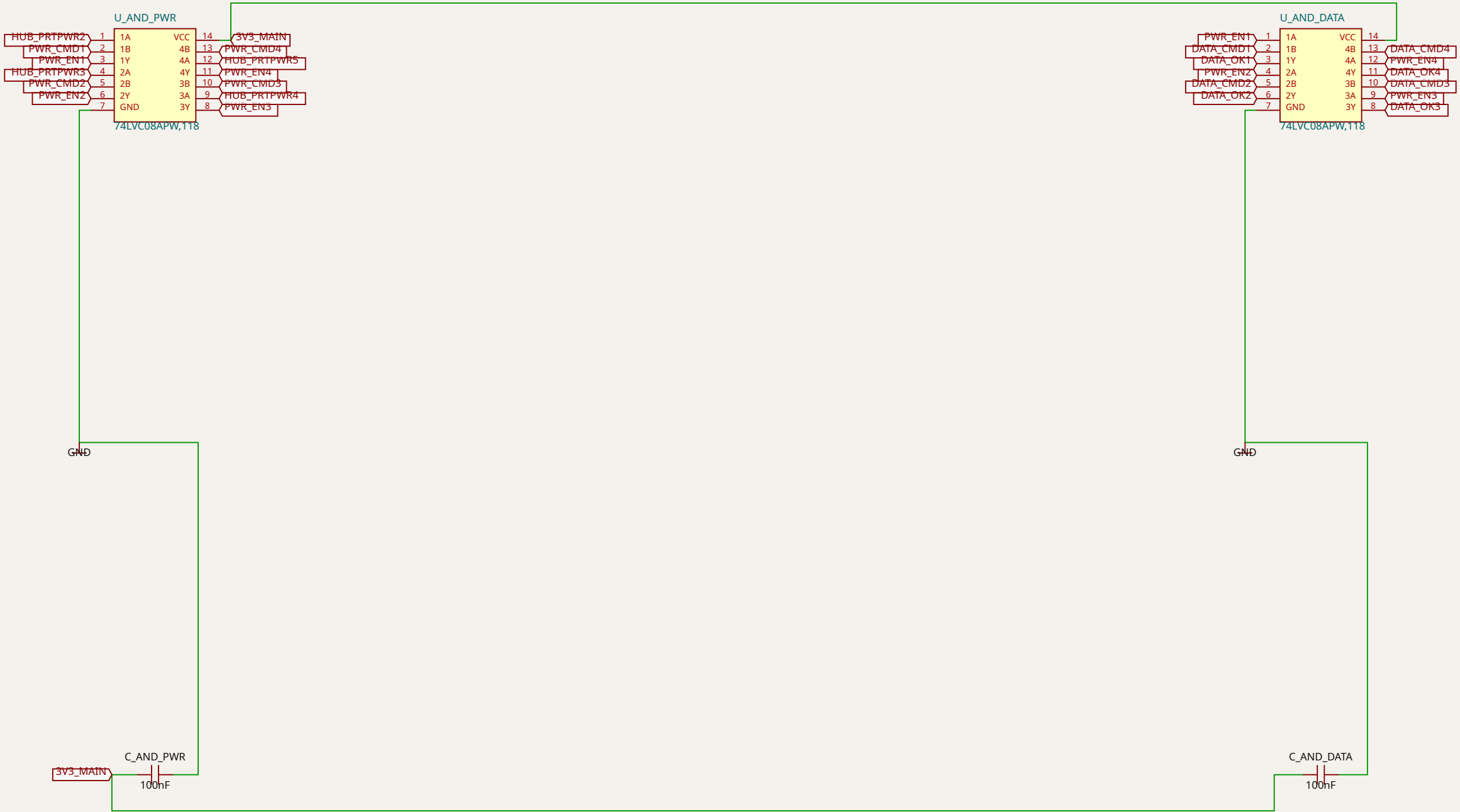
Page 5 of 10 • exact circuit.json SHA-256 91fa12ab65d3eac8... • 22 components • landscape page fit



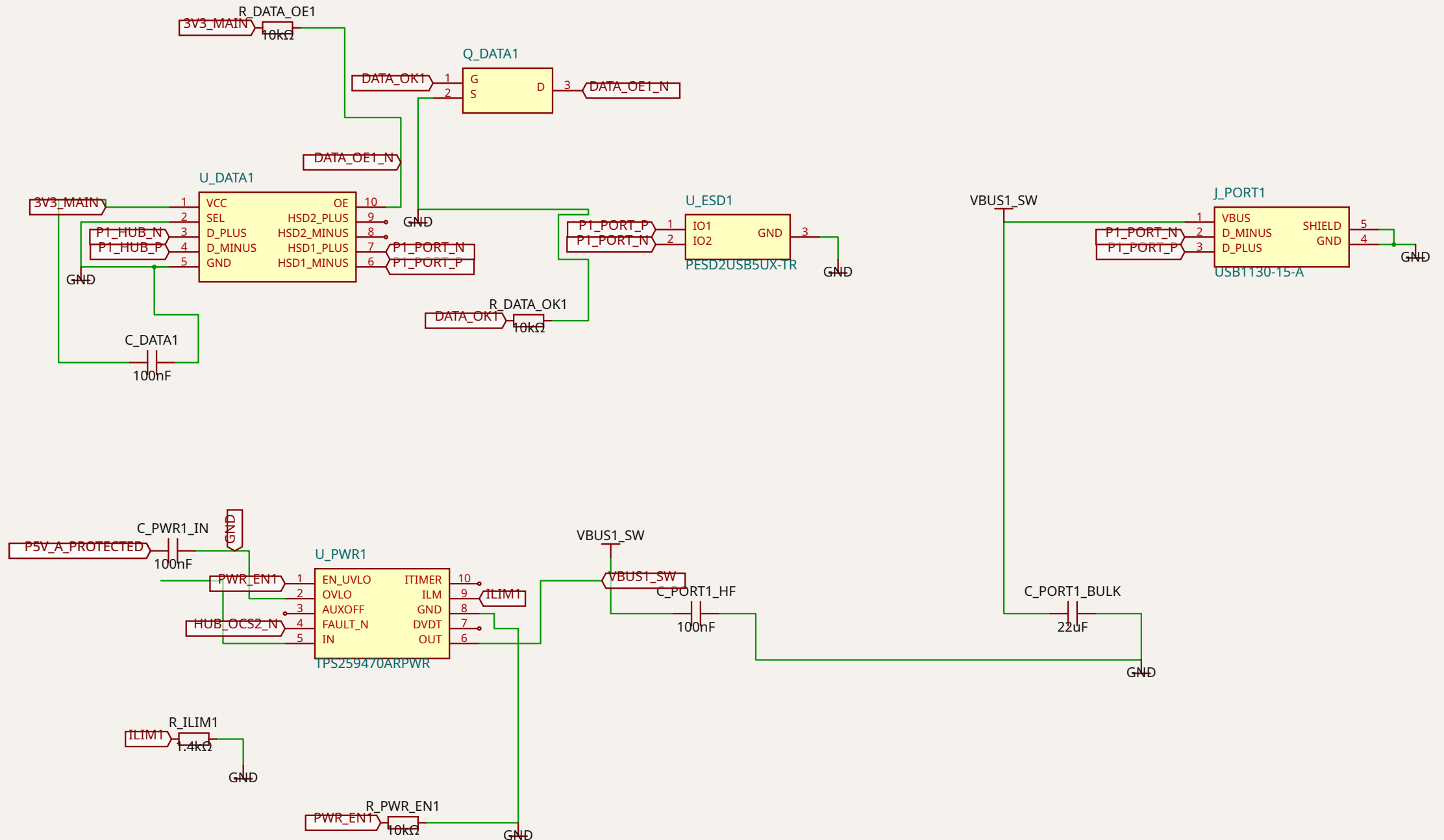
INTENTIONAL NC: unused U_CTRL GPIO/UART; U_EXP GPB/INT/NC pins.

USB_CONTROLLED_DEBUG_HUB_2A

INTERLOCKS — hub policy AND host command; data follows commanded power enable

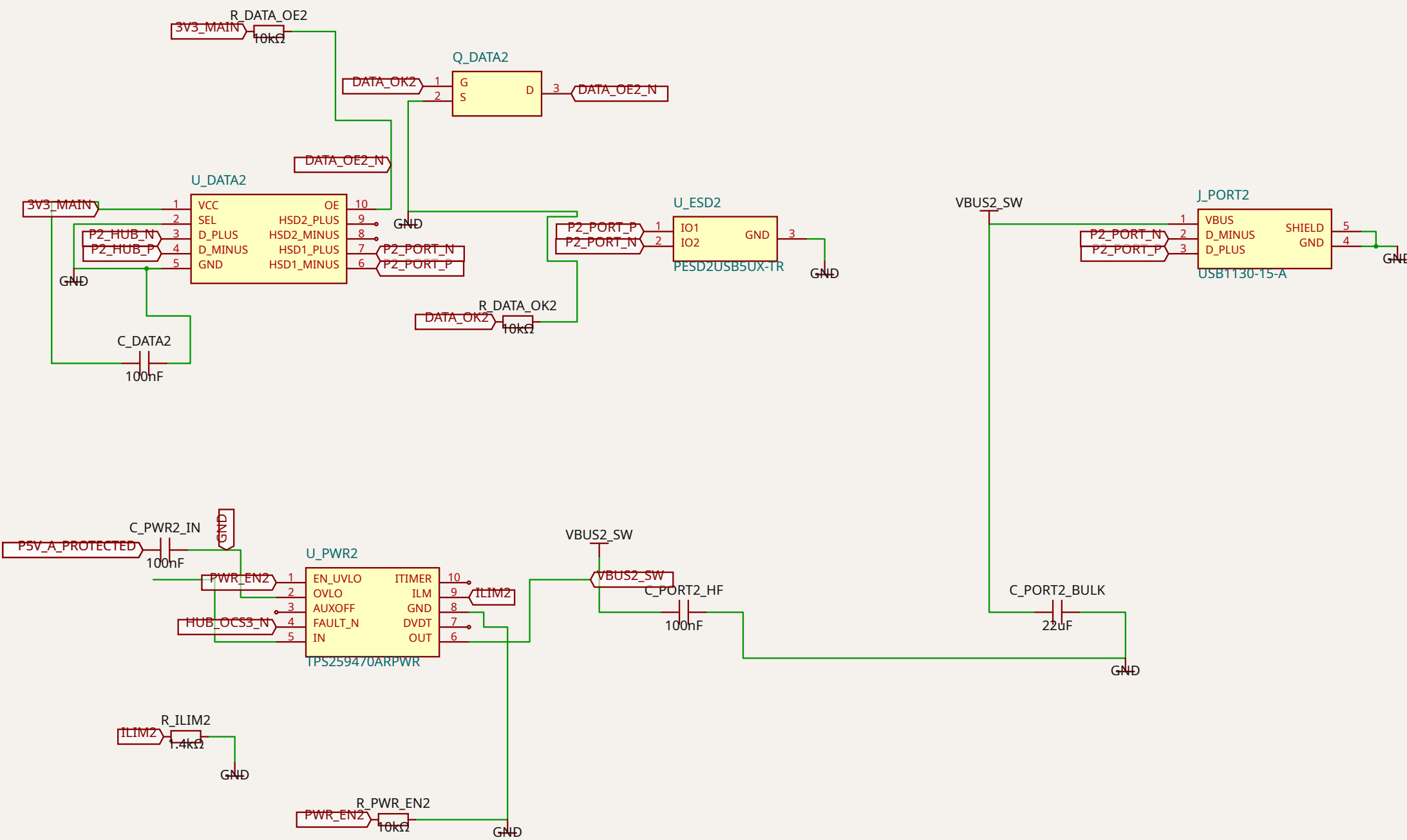


EXTERNAL PORT 1 — independent power/data disconnect



USB_CONTROLLED_DEBUG_HUB_2A

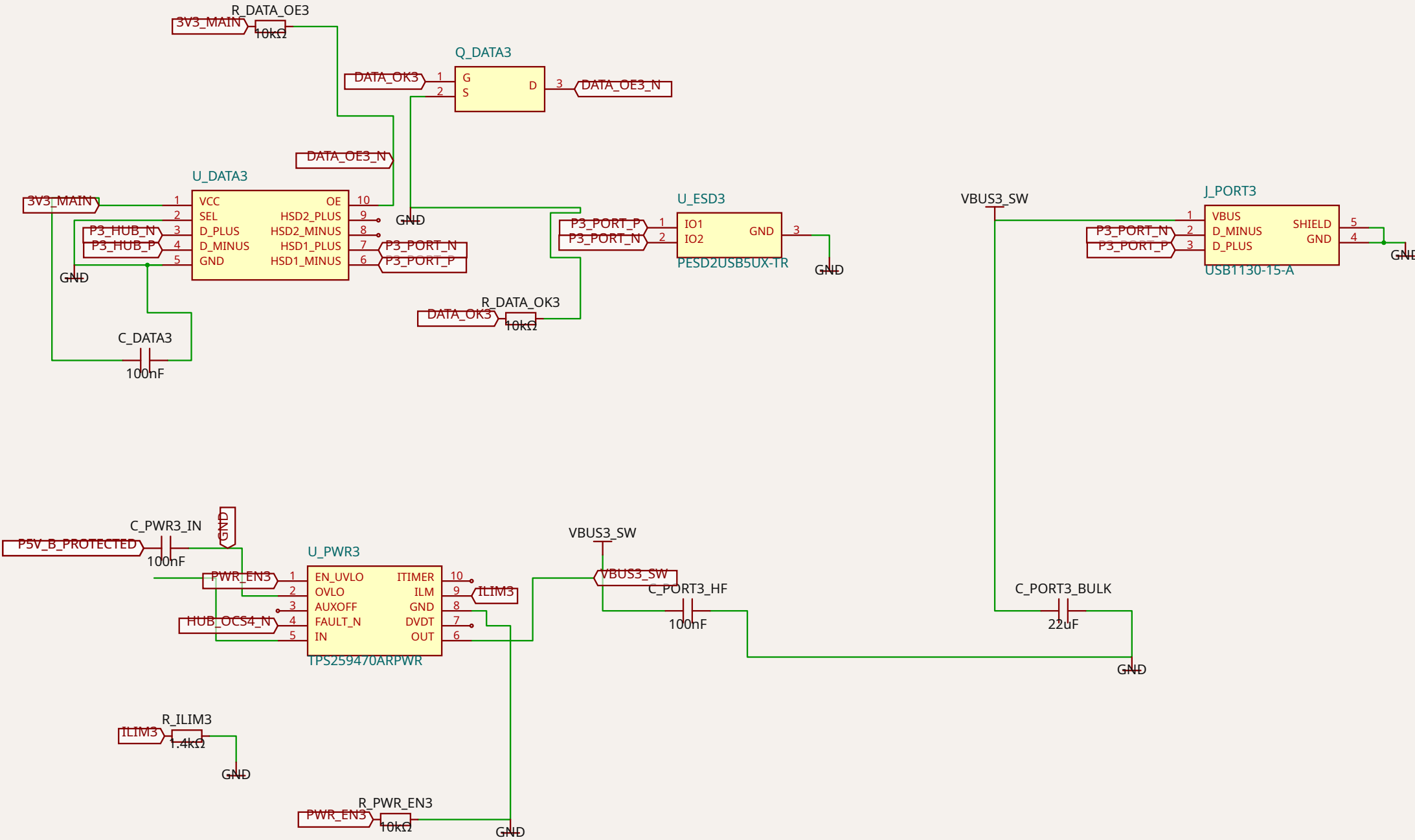
EXTERNAL PORT 2 — independent power/data disconnect



USB_CONTROLLED_DEBUG_HUB_2A

EXTERNAL PORT 3 — independent power/data disconnect

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USB_CONTROLLED_DEBUG_HUB_2A

EXTERNAL PORT 4 — independent power/data disconnect

